

Security Concepts

Module 1:

Unit 1: The Importance of cyber

Security



Learning Objective

LO1

Define cyber security concepts using practical examples

LO2

Discuss the insecurities of cyber space

LO3

Explain the importance of the concepts of data information and knowledge in the context of cyber security

LO5

Describe how authentication and authorization is used in securing data, information and applications

LO4

outline the role of confidentiality, integrity, availability, and non-repudiation in a given security problem

LO6

Discuss information security domains using examples

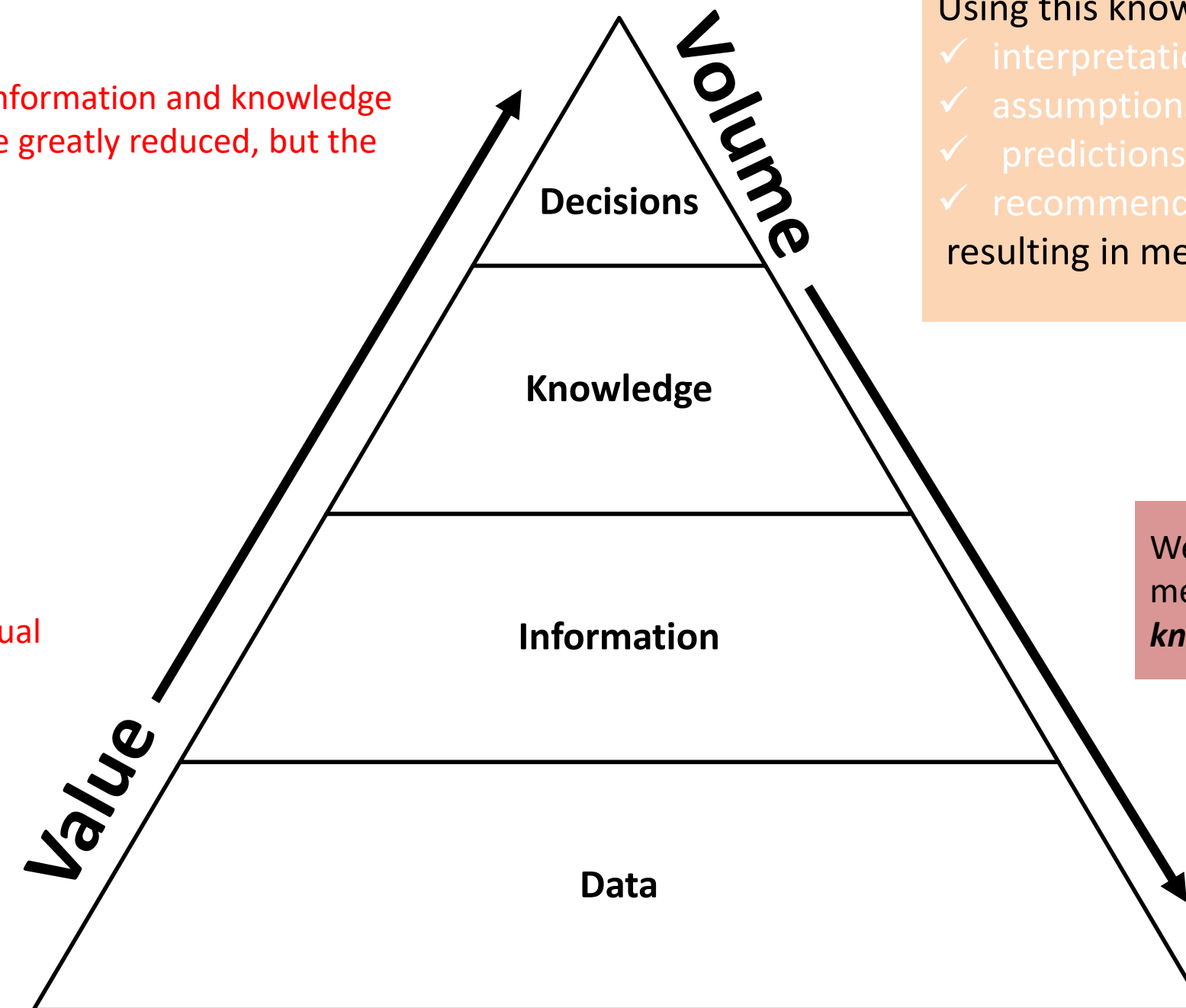


The Knowledge Hierarchy

The Journey of Data

As we move up the scale, through information and knowledge to decision-making, the volumes are greatly reduced, but the value increases dramatically.

But the value of each individual item of data is low



Using this knowledge, we can make:

- ✓ interpretations
 - ✓ assumptions
 - ✓ predictions
 - ✓ recommendations
- resulting in meaningful **decisions**

We then combine these Facts and figures to construct meaningful **information** that informs us and provides **knowledge**.

At the Data Level, Volumes are high

We begin the journey by Collecting **Data** :

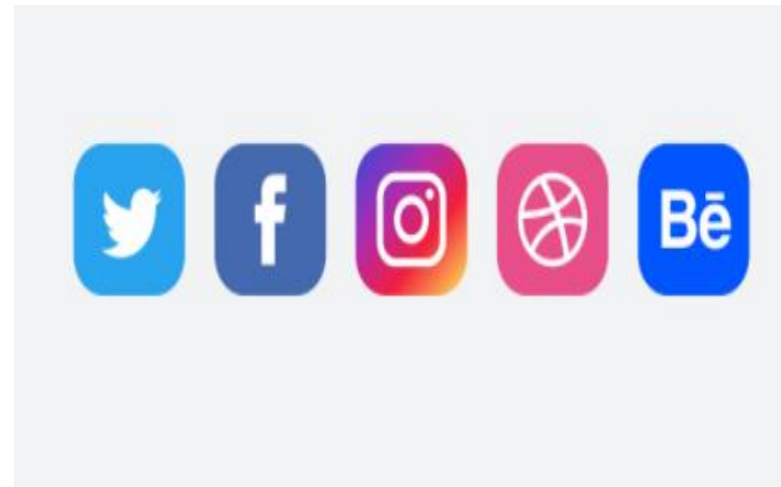
- ✓ Facts
- ✓ Figures

Sources of Data

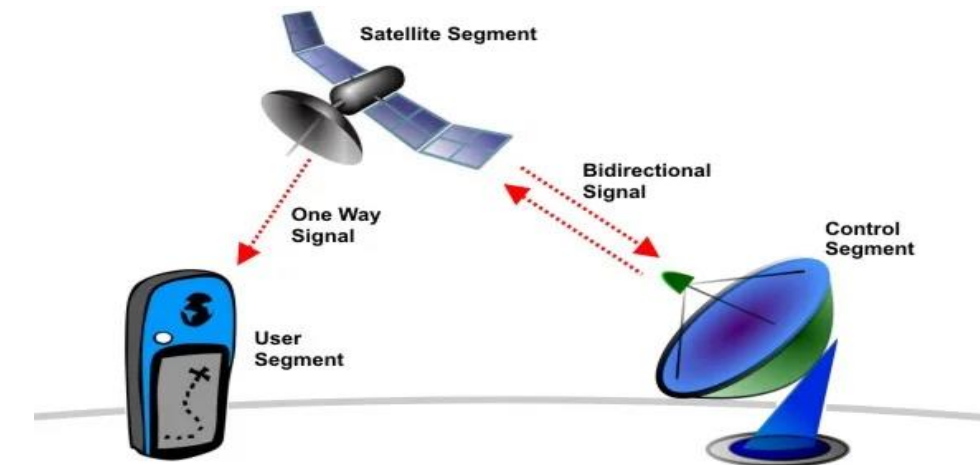
Call Data Records



Social Networks



Global Positioning System: GPS



Fitness Tracking Data



Credit and Debit Card Transactions



Number Plate Recognition Data



What could be the Implication?

Hackers would know a great deal about you:

- Movements
- Relationships,
- both business and personal
- Spending habits,
- Religion,
- Sexual orientation and/or gender,
- Political views and your
- General health.

The Critical Information Security Issues

Three Fundamental Terms of Information Security

Confidentiality: Information is not disclosed to unauthorized people



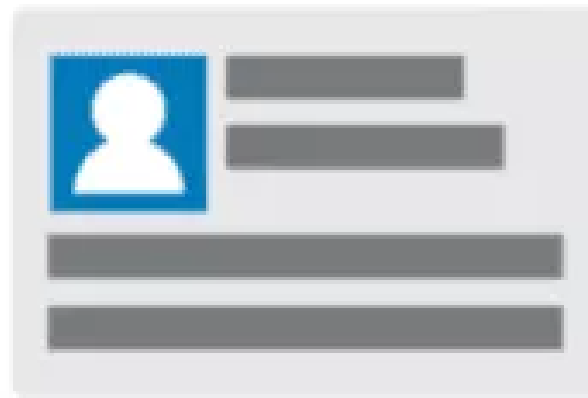
Integrity: Information is not changed or altered at **rest** or in **transit**

Availability: Information is available for use when required

The Critical Information Security Issues

Other Terms of Information Security

Authentication: Identifies the user's Identity- Are you who you say you are?

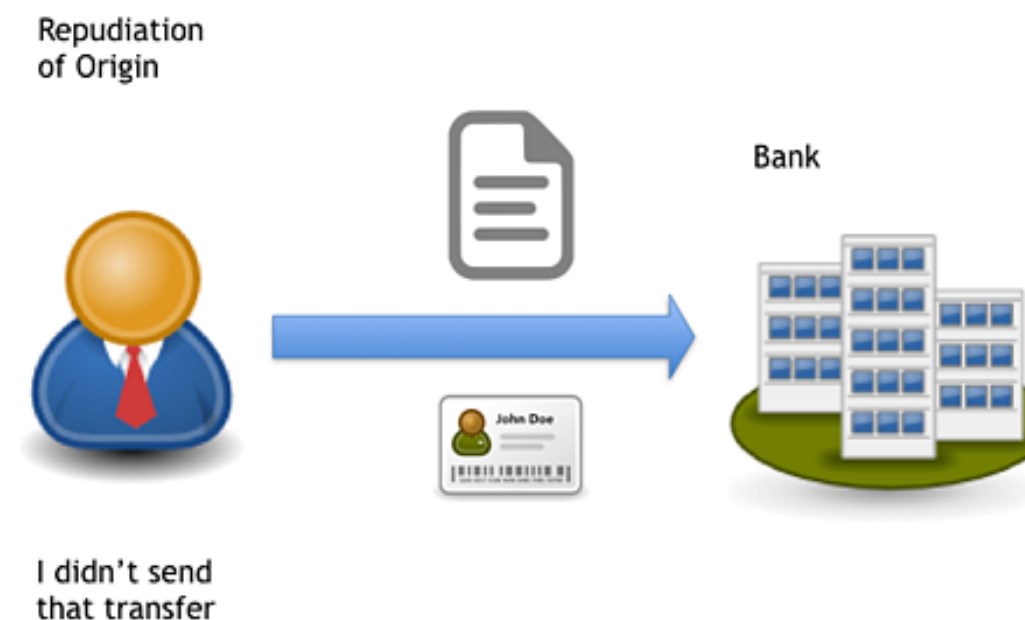


Authorization: After authenticated, what are you allowed to do?



VS

Non-repudiation : Authenticated users cannot deny any actions carried out a particular action



The Critical Information Security Issues

Other Terms of Information Security (Conti..)

Privacy: Brings in a more personal view, is much more personal to us.

VS

Security : Something is protected from unauthorized access or harm

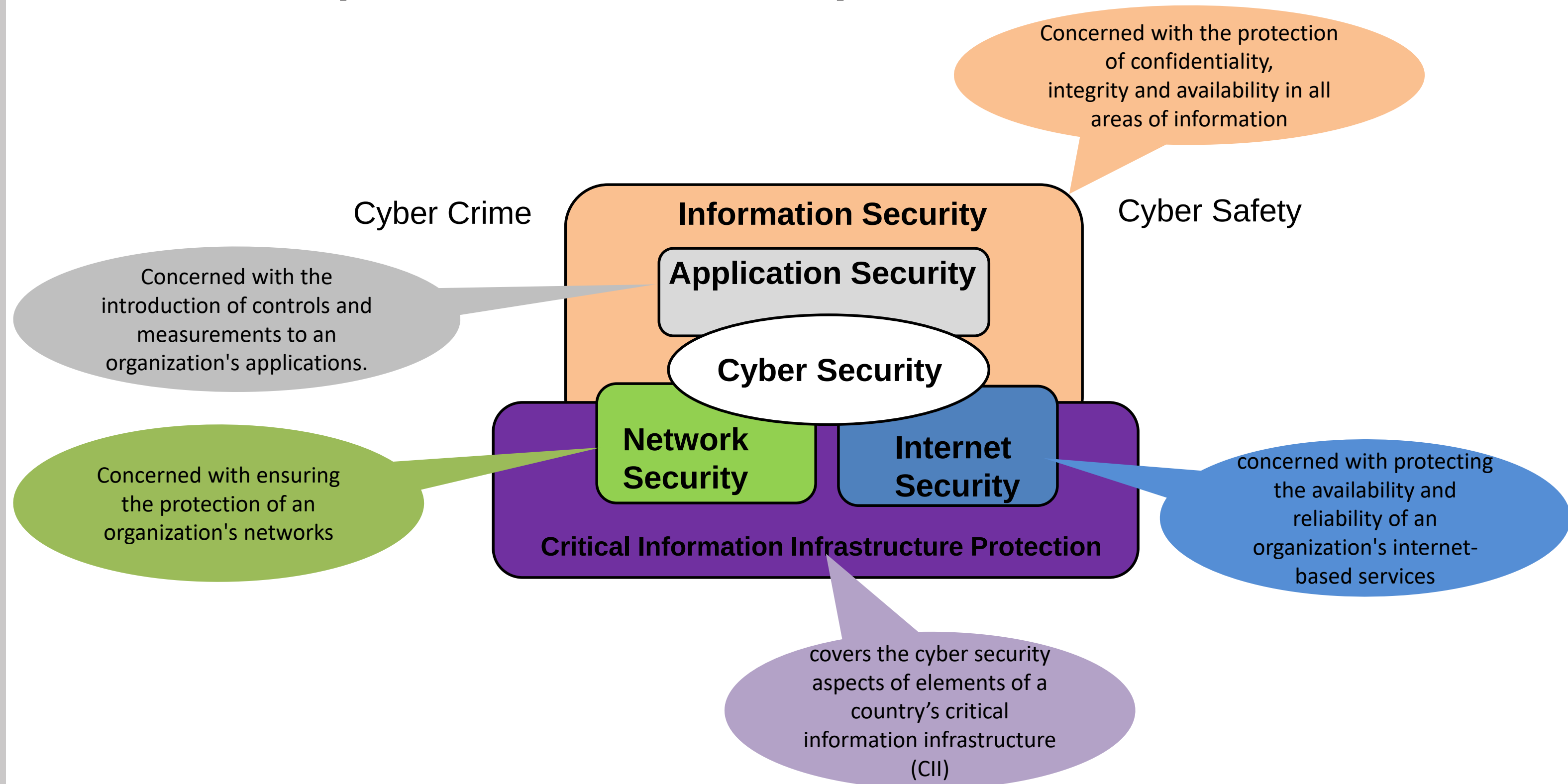


The Expectation of Users from Organization

- **Users :** Expect that their data will not be impacted by the issues described on the previous slides.
- Organization handling information about them will treat it in confidence and with respect.
- In accordance with the appropriate legislation such as the Data Protection Act (DPA),
- Will not allow it to be made available to those who have no entitlement to see it.
- This includes the 'selling on' of users' information, much of which is completely illegal.

Cyber Security in the Wider Context

Relationships between security domains



PREPARE FOR A
PRODUCTIVE YEAR

